Summary of Calculation Logic for Raman Detuning and Velocity

Physical Background

In cold atom Raman two-photon transitions, the resonance condition is influenced by atomic velocity, Doppler shifts, laser geometry, and the two-photon recoil shift. This program computes the detuning δ (in kHz) or the transverse velocity v_x (in mm/s) according to user input and experimental settings. It also considers two possible hyperfine transitions: F1→F2 and F2→F1.

Symbols and Parameters

- v_z : Vertical (longitudinal) velocity component (m/s, fixed constant)
- v_x : Horizontal (transverse) velocity component (m/s, or mm/s as user input/output)
- α : Angle between the Raman laser and the z-axis (degrees, fixed constant)
- k_{eff} : Effective two-photon wavevector, $k_{\text{eff}} = 2 \times \frac{2\pi}{\lambda_{\text{laser}}}$
- ω_r : Two-photon recoil frequency (rad/s, fixed constant)
- δ : Raman detuning (user input or to be calculated, in kHz)
- Transition type: F1→F2 or F2→F1

Detuning Calculation Formulas (Given v_x, Compute δ)

$$\delta = k_{\text{eff}} v_z \sin\alpha \pm k_{\text{eff}} v_x \cos\alpha \pm \omega_r$$

The sign in front of ω_r depends on the transition direction:

- F1→F2: $+\omega_r$
- F2→F1: $-\omega_r$

The explicit formulas for the four physical cases are:

$$\begin{aligned}
\text{flying up, } \Delta > 0: &\quad + k_{\text{eff}} v_z \sin\alpha + k_{\text{eff}} v_x \cos\alpha + s\omega_r \\
\text{flying up, } \Delta < 0: &\quad - k_{\text{eff}} v_z \sin\alpha - k_{\text{eff}} v_x \cos\alpha + s\omega_r \\
\text{falling down, } \Delta > 0: &\quad + k_{\text{eff}} v_z \sin\alpha - k_{\text{eff}} v_x \cos\alpha + s\omega_r \\
\text{falling down, } \Delta < 0: &\quad - k_{\text{eff}} v_z \sin\alpha + k_{\text{eff}} v_x \cos\alpha + s\omega_r
\end{aligned}$$

where $s = +1$ for F1→F2 and $s = -1$ for F2→F1.

Velocity Inversion Formulas (Given δ, Compute v_x)

Let δ' be the detuning in rad/s, i.e., $\delta' = 2\pi \times 1000 \times \delta$.

- **Flying up**

$$\begin{aligned} \text{If } \delta \geq 0: \quad v_x &= \frac{\delta' - k_{\text{eff}} v_z \sin\alpha - s\omega_r}{k_{\text{eff}} \cos\alpha} \\ \text{If } \delta < 0: \quad v_x &= -\frac{\delta' - s\omega_r + k_{\text{eff}} v_z \sin\alpha}{k_{\text{eff}} \cos\alpha} \end{aligned}$$

- **Falling down**

$$\begin{aligned} \text{If } \delta \geq 0: \quad v_x &= -\frac{\delta' - k_{\text{eff}} v_z \sin\alpha - s\omega_r}{k_{\text{eff}} \cos\alpha} \\ \text{If } \delta < 0: \quad v_x &= \frac{\delta' - s\omega_r + k_{\text{eff}} v_z \sin\alpha}{k_{\text{eff}} \cos\alpha} \end{aligned}$$

where again $s = +1$ for F1→F2 and $s = -1$ for F2→F1.

Program Logic

1. The user selects the mode (vx → detuning or detuning → vx), enters the value, chooses flying up/falling down, and specifies the transition type (F1→F2 or F2→F1).
2. The program automatically selects the correct formula based on user input and, for the velocity inversion, also automatically determines the sign of δ.
3. The result and the formula used are shown to the user.

Note

All terms use SI units, except where user input/output is in mm/s (for v_x) or kHz (for δ).